FUTURE INTERNS

CYBERSECURITY TASK-3
API SECURITY RISK ANALYSIS
(MODERN SAAS SKILL)

PREPARED BY

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INTRODUCTION

Today, most apps and websites use APIs to share data and provide services. APIs often handle important information like user details and login data, so they are a common target for security attacks.

If an API is not secured properly, attackers can access sensitive data or misuse the system. The aim of this task is to perform a read-only API Security Risk Analysis on a public test API. The API is reviewed by testing endpoints and checking responses to identify common security risks and suggest simple improvements, without attacking or exploiting the system.

API SECURITY AND TOOLS

API Security

API security is about protecting APIs from unauthorized access and data leaks. Since APIs are widely used and often handle user data, they must be checked for common security risks.

Target API

The target API for this analysis is JSONPlaceholder, a public test API used for learning and testing. It provides sample endpoints like users and posts and is safe to use for educational security analysis
https://jsonplaceholder.typicode.com.

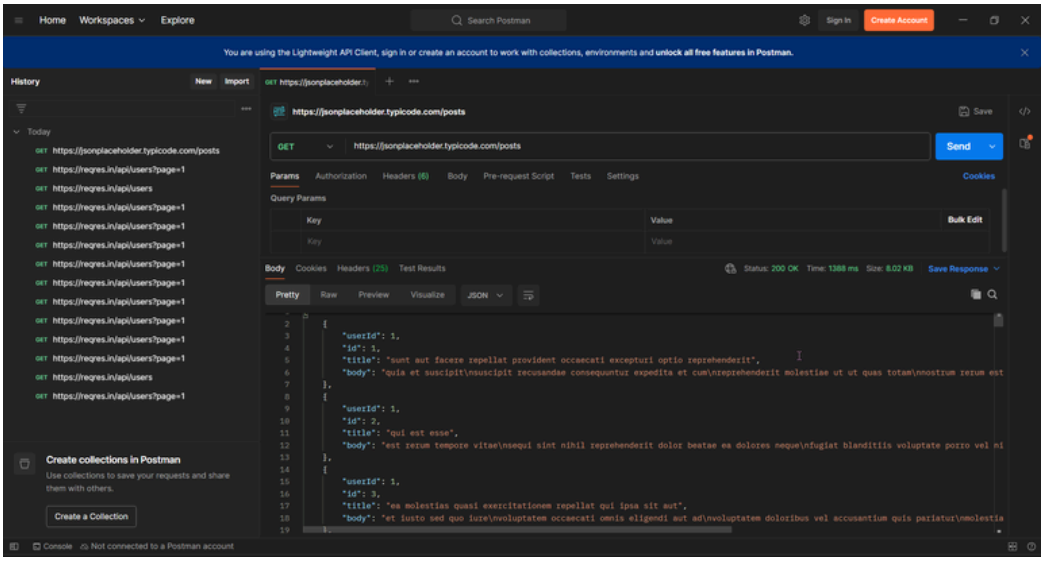
Tools Used

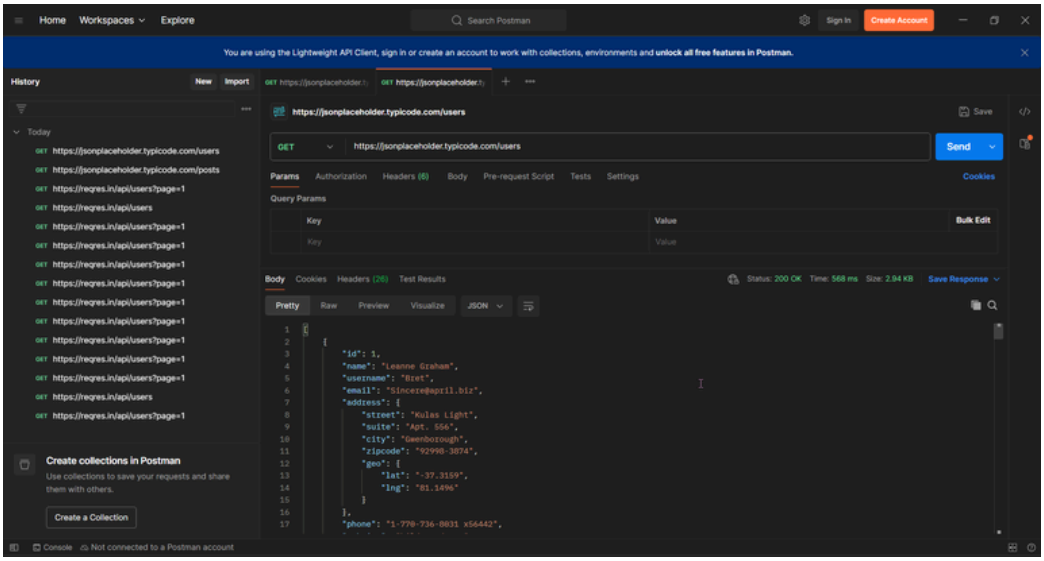
Postman and Insomnia – To send API requests and check responses.
GitHub – To store and share the project files.

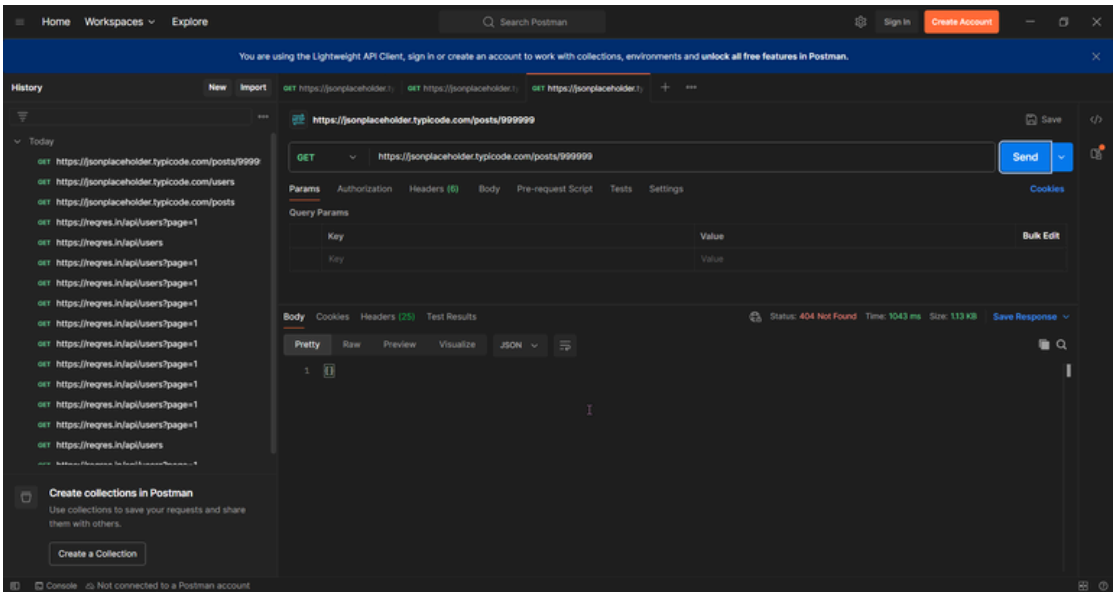
POSTMAN TOOL

Postman is used to send API requests and view responses like status codes and data. In this project, it helped identify risks such as open access to endpoints and lack of visible security controls.

Using postman tool for target API:







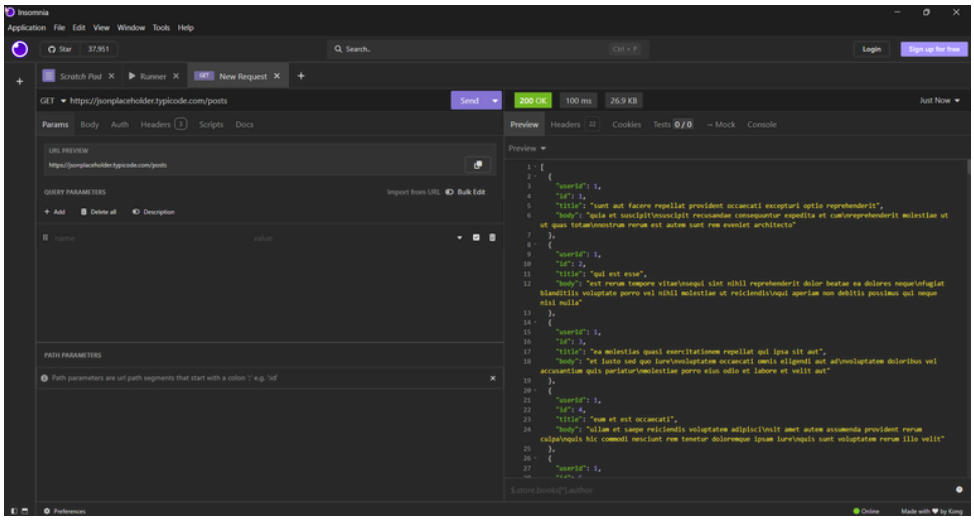
Results and Findings

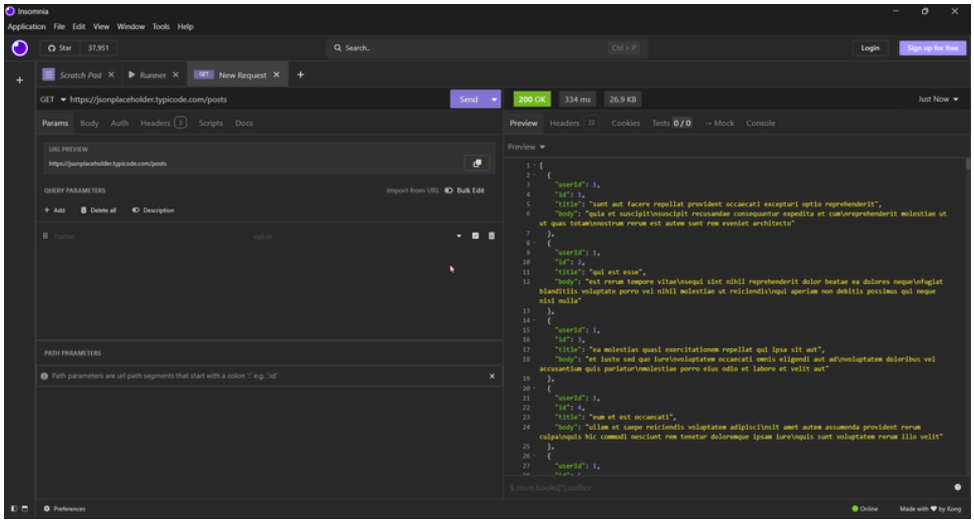
- The API endpoints can be accessed without authentication.
- User data such as name, email, phone, and address is returned openly.
- This shows excessive data exposure in API responses.
- No visible rate-limiting or access control headers were found.
- Error handling is basic and does not clearly restrict invalid requests.
- These issues can lead to data misuse or large-scale data scraping in real systems.

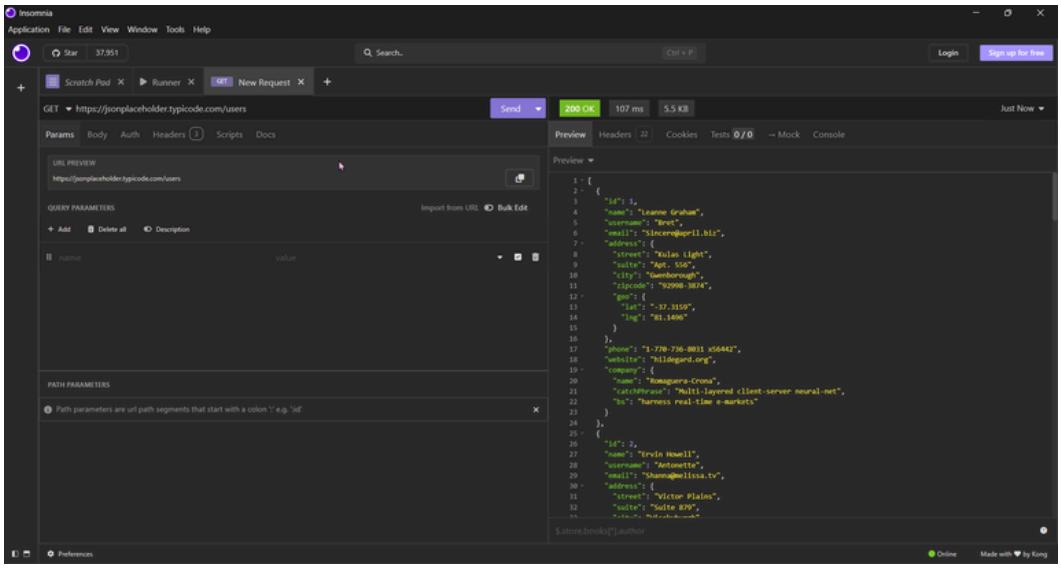
INSOMNIA TOOL

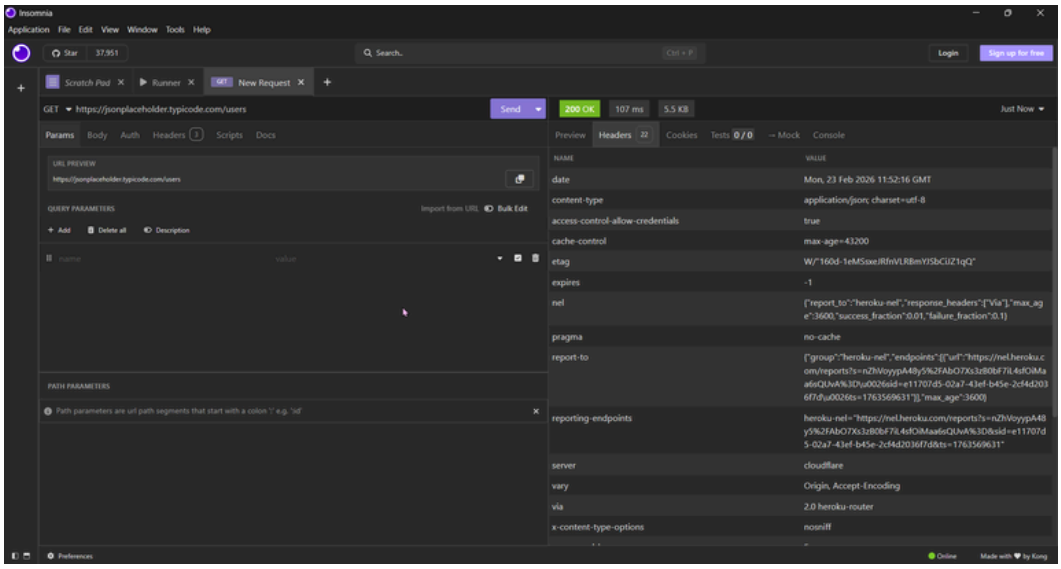
Insomnia is an API testing tool used to send requests and view responses in a simple interface. In this project, Insomnia was used to test public API endpoints and observe the returned data and headers. It helped confirm that the API allows open access to data without authentication.

Using insomnia tool for target API:









Results and Findings

- The API endpoints were accessible without any authentication.
- Sensitive user information was returned in the responses.
- No visible rate-limiting or access control headers were found.
- Error handling was basic and did not restrict invalid requests.